

**E.G.S. PILLAY ENGINEERING COLLEGE, (Autonomous)**

Approved by AICTE, New Delhi

Affiliated to Anna University, Chennai | Accredited by NAAC with „A++

Grade Accredited by NBA Tier-1



NAGAPATTINAM – 611002

**B.E.–Electrical and Electronics Engineering**

**R-2023 - SEMESTER IV**

| SEMESTER IV                  |             |                                           |    |    |    |    |               |     |       |          |       |
|------------------------------|-------------|-------------------------------------------|----|----|----|----|---------------|-----|-------|----------|-------|
| Course Code                  | Course Name | L                                         | T  | P  | H  | C  | Maximum Marks |     |       | Category |       |
|                              |             |                                           |    |    |    |    | CA            | ES  | Total |          |       |
| Theory Course                |             |                                           |    |    |    |    |               |     |       |          |       |
| 1                            | 2302EE401   | Generation, Transmission and Distribution | 3  | 2  | 0  | 5  | 4             | 40  | 60    | 100      | PCC   |
| 2                            | 2302EE402   | Electrical Machines -II                   | 3  | 2  | 0  | 5  | 4             | 40  | 60    | 100      | PCC   |
| 3                            | 2302EE403   | Power Electronics                         | 3  | 0  | 0  | 3  | 3             | 40  | 60    | 100      | PCC   |
| 4                            | 2302EE404   | Linear Control Systems                    | 3  | 2  | 0  | 5  | 4             | 40  | 60    | 100      | PCC   |
| 5                            | 2301MC401   | Mandatory course-I                        | 2  | 0  | 0  | 2  | 0             | 100 | 0     | 100      | MC    |
| Theory Cum Laboratory Course |             |                                           |    |    |    |    |               |     |       |          |       |
| 6                            | 2302EE405   | Linear Integrated Circuits                | 3  | 0  | 2  | 5  | 4             | 50  | 50    | 100      | PCC   |
| 7                            | 2301HSX01   | Universal Human Values and Ethics         | 1  | 0  | 2  | 3  | 2             | 50  | 50    | 100      | HSSMC |
| Practical Course             |             |                                           |    |    |    |    |               |     |       |          |       |
| 8                            | 2302EE451   | Electrical Machines –II Laboratory        | 0  | 0  | 2  | 2  | 1             | 60  | 40    | 100      | PCC   |
| 9                            | 2302EE452   | Power Electronics and Drives Laboratory   | 0  | 0  | 2  | 2  | 1             | 60  | 40    | 100      | PCC   |
| 10                           | 2304GE401   | Professional Development -2               | 0  | 0  | 2  | 2  | 1             | 100 | 0     | 100      | EEC   |
| 11                           | 2301LS401   | Life Skill :04                            | 0  | 0  | 0  | 0  | 0             | 100 | 0     | 100      | -     |
| Total                        |             |                                           | 18 | 06 | 10 | 34 | 24            | 680 | 420   | 1100     |       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                           |                                                                                                          |      |     |     |     |     |     |     |     |     |          |   |   |
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| 2302EE401                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | GENERATION, TRANSMISSION AND DISTRIBUTION |                                                                                                          |      |     |     |     |     |     |     |     | L   | T        | P | C |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                           |                                                                                                          |      |     |     |     |     |     |     |     | 3   | 2        | 0 | 4 |
| PREREQUISITE:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                           |                                                                                                          |      |     |     |     |     |     |     |     |     |          |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                           |                                                                                                          |      |     |     |     |     |     |     |     |     |          |   |   |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                           | Electric Circuit Analysis                                                                                |      |     |     |     |     |     |     |     |     |          |   |   |
| COURSE OBJECTIVES:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                           |                                                                                                          |      |     |     |     |     |     |     |     |     |          |   |   |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                           | To describe the principle of operation of generation, transmission and distribution system.              |      |     |     |     |     |     |     |     |     |          |   |   |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                           | To learn about different insulators and underground cables                                               |      |     |     |     |     |     |     |     |     |          |   |   |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                           | To study the line parameters and interference with neighboring circuits.                                 |      |     |     |     |     |     |     |     |     |          |   |   |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                           | To understand the mechanical design and performance analysis of transmission lines.                      |      |     |     |     |     |     |     |     |     |          |   |   |
| COURSE OUTCOMES:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                           |                                                                                                          |      |     |     |     |     |     |     |     |     |          |   |   |
| On the successful completion of the course, students will be able to                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                           |                                                                                                          |      |     |     |     |     |     |     |     |     |          |   |   |
| CO1:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                           | Discuss the structure of power system, power generation from renewable and non-renewable energy sources  |      |     |     |     |     |     |     |     |     |          |   |   |
| CO2:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                           | Calculate the sag and tension of overhead lines and string efficiency of insulators                      |      |     |     |     |     |     |     |     |     |          |   |   |
| CO3:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                           | Compute the parameters of transmission lines, and underground cables                                     |      |     |     |     |     |     |     |     |     |          |   |   |
| CO4:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                           | Calculate the transmission efficiency and voltage regulation of a transmission line                      |      |     |     |     |     |     |     |     |     |          |   |   |
| CO5:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                           | Explain the distribution systems, substation and modern trends in transmission and distribution systems. |      |     |     |     |     |     |     |     |     |          |   |   |
| COs Vs. POs MAPPING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                           |                                                                                                          |      |     |     |     |     |     |     |     |     |          |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | COs                                       | PO1                                                                                                      | PO2  | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | P10 | P11      |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO1                                       | 2                                                                                                        | 1    | 1   |     |     |     |     |     |     |     |          |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO2                                       | 3                                                                                                        | 2    | 1   |     |     |     |     |     |     |     |          |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO3                                       | 3                                                                                                        | 2    | 1   |     |     |     |     |     |     |     |          |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO4                                       | 3                                                                                                        | 2    | 1   |     |     |     |     |     |     |     |          |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO5                                       | 2                                                                                                        | 1    | 1   |     |     |     |     |     |     |     |          |   |   |
| COs Vs. PSOs MAPPING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                           |                                                                                                          |      |     |     |     |     |     |     |     |     |          |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | COs                                       | PSO1                                                                                                     | PSO2 |     |     |     |     |     |     |     |     |          |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO1                                       | 3                                                                                                        |      |     |     |     |     |     |     |     |     |          |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO2                                       | 3                                                                                                        |      |     |     |     |     |     |     |     |     |          |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO3                                       | 3                                                                                                        |      |     |     |     |     |     |     |     |     |          |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO4                                       | 3                                                                                                        |      |     |     |     |     |     |     |     |     |          |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO5                                       | 3                                                                                                        |      |     |     |     |     |     |     |     |     |          |   |   |
| COURSE CONTENTS:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                           |                                                                                                          |      |     |     |     |     |     |     |     |     |          |   |   |
| MODULE I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           | ELECTRICAL POWER GENERATION<br>(Block diagram/Qualitative approach only)                                 |      |     |     |     |     |     |     |     |     | 12 Hours |   |   |
| Structure of electric power system: Single line diagram- different operating voltages of generation, transmission and distribution- transmission line route survey (Marking diagrams in India map).Types of energy and classification of power plants; Present power position in India, future planning for power generation.<br>Power generation from Non-Renewable energy sources: Thermal and Nuclear based power generation<br>Power generation from Renewable energy sources: Solar, Wind, Hydro, Tidal, Geothermal, Fuel cell and Bio-mass based power generation. |                                           |                                                                                                          |      |     |     |     |     |     |     |     |     |          |   |   |
| MODULE II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                           | MECHANICAL DESIGN OF LINES, AND INSULATORS                                                               |      |     |     |     |     |     |     |     |     | 12 Hours |   |   |

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| <b>Mechanical design of OH lines:</b> Line supports -types of towers - tension and sag calculation - towers at equal heights, unequal heights- effects of wind and ice loading- stringing chart; Methods of grounding.<br><b>Insulators:</b> Types - voltage distribution in insulator string - improvement of string efficiency - testing of insulators.                                                                                                                                                                                                                                                                                                                                                                                              |                                                                 |                 |
| <b>MODULE III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>PARAMETERS OF TRANSMISSION LINES, AND UNDERGROUND CABLES</b> | <b>12 Hours</b> |
| <b>Parameters of single and three phase transmission lines:</b> Resistance, inductance of transmission lines- inductance of a single phase two wire line, inductance of three phase lines with symmetrical and unsymmetrical spacing; Capacitance of a single phase two wire line, capacitance of a three phase line with symmetrical and unsymmetrical spacing, transposed, solid, stranded and bundled conductors; selfand mutual GMD.<br><b>Underground cables:</b> Constructional features of LT and HT cables; Capacitance of single-core cable, capacitance of 3-core belted cable; Insulation resistance; potential gradient - grading of cables; power factor and heating; DC cables- Cable standards and procedures-Cable faults and testing. |                                                                 |                 |
| <b>MODULE IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>PERFORMANCE OF TRANSMISSION LINES</b>                        | <b>12 Hours</b> |
| Performance of Transmission lines - short line, medium line and long line, equivalent circuits- Phasor diagram- attenuation constant- phase constant-surge impedance; Calculation of transmission efficiency and voltage regulation; Real and reactive power flow in lines; Surge impedance loading; Power circle diagrams; Skin and proximity effects, Ferranti effect; Interference with neighboring communication circuits; Corona discharge characteristics - Critical voltage and corona loss                                                                                                                                                                                                                                                     |                                                                 |                 |
| <b>MODULE V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>DISTRIBUTION SYSTEMS AND SUBSTATION</b>                      | <b>12 Hours</b> |
| <b>Distribution systems:</b> General Aspects-AC and DC distribution; Radial and ring main systems; Calculation of voltage in distributors with concentrated and distributed loads, Kelvin's law; Techniques of voltage control and power factor improvement.<br><b>Substation:</b> Types- typical key diagram of an 11kV / 400V substation; Grounding; Recent trends in transmission and distribution: EHVAC, HVDC, GIS and FACTS (Qualitative treatment only).                                                                                                                                                                                                                                                                                        |                                                                 |                 |
| <b>TOTAL: 60 HOURS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                 |                 |
| <b>REFERENCES:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                 |                 |
| 1. D.P.Kothari, I.J. Nagarath, 'Power System Engineering', Mc Graw-Hill Publishing Company limited, New Delhi, Third Edition, 2019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                 |                 |
| 2. C.L.Wadhwa, 'Electrical Power Systems', New Age International Ltd, seventh edition 2022                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                 |                 |
| 3. S.N. Singh, 'Electric Power Generation, Transmission and Distribution', Prentice Hall of India Pvt. Ltd, New Delhi, Second Edition, 2008                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                 |                 |
| 4. B.R.Gupta, 'Power System Analysis and Design' S. Chand, New Delhi, Sixth Edition, 2011                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                 |                 |
| 5. Hadi Saadat, 'Power System Analysis', McGraw Hill Education Pvt. Ltd., New Delhi, 3rd Edition, 23 <sup>rd</sup> reprint, 2015                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                 |                 |
| 6. R.K.Rajput, 'A Text Book of Power System Engineering' 2 <sup>nd</sup> edition, Laxmi Publications (P) Ltd, New Delhi, 2016                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                 |                 |
| 7. V.K.Mehta, Rohit Mehta, 'Principles of power system', S. Chand & Company Ltd, New Delhi, 2013.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                 |                 |
| 8. <a href="https://nptel.ac.in/courses/108102047">https://nptel.ac.in/courses/108102047</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                 |                 |
| 9. <a href="https://nptel.ac.in/courses/108108099">https://nptel.ac.in/courses/108108099</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                 |                 |

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|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 2302EE402                                                            | ELECTRICAL MACHINES - II                                                                                    | L    | T    | P   | C   |     |     |     |     |     |     |     |
|                                                                      |                                                                                                             | 3    | 2    | 0   | 4   |     |     |     |     |     |     |     |
| PREREQUISITE:                                                        |                                                                                                             |      |      |     |     |     |     |     |     |     |     |     |
|                                                                      |                                                                                                             |      |      |     |     |     |     |     |     |     |     |     |
|                                                                      | 1. Electric Circuit Analysis                                                                                |      |      |     |     |     |     |     |     |     |     |     |
|                                                                      | 2.Electrical Machines -I                                                                                    |      |      |     |     |     |     |     |     |     |     |     |
|                                                                      |                                                                                                             |      |      |     |     |     |     |     |     |     |     |     |
| COURSE OBJECTIVES:                                                   |                                                                                                             |      |      |     |     |     |     |     |     |     |     |     |
|                                                                      | 1. To impart the basic operation and construction of various AC machines.                                   |      |      |     |     |     |     |     |     |     |     |     |
|                                                                      | 2. To describe the performance of synchronous machine by different methods.                                 |      |      |     |     |     |     |     |     |     |     |     |
|                                                                      | 3. To analyze the performance characteristics and equivalent circuits of AC machines.                       |      |      |     |     |     |     |     |     |     |     |     |
|                                                                      |                                                                                                             |      |      |     |     |     |     |     |     |     |     |     |
| COURSE OUTCOMES:                                                     |                                                                                                             |      |      |     |     |     |     |     |     |     |     |     |
|                                                                      |                                                                                                             |      |      |     |     |     |     |     |     |     |     |     |
| On the successful completion of the course, students will be able to |                                                                                                             |      |      |     |     |     |     |     |     |     |     |     |
| CO1:                                                                 | Investigate the percentage regulation of three-phase AC generator using various regulation methods.         |      |      |     |     |     |     |     |     |     |     |     |
| CO2:                                                                 | Inspect the performance characteristics of three-phase synchronous motor by conducting various test         |      |      |     |     |     |     |     |     |     |     |     |
| CO3:                                                                 | Identify the performance characteristics of three-phase induction motor by conducting OC and SC test        |      |      |     |     |     |     |     |     |     |     |     |
| CO4:                                                                 | Gain Knowledge about the concepts of starters & speed control methods                                       |      |      |     |     |     |     |     |     |     |     |     |
| CO5:                                                                 | Describe the characteristics behavior of various types of single-phase induction motor and special machines |      |      |     |     |     |     |     |     |     |     |     |
|                                                                      |                                                                                                             |      |      |     |     |     |     |     |     |     |     |     |
| COs Vs. POs MAPPING                                                  |                                                                                                             |      |      |     |     |     |     |     |     |     |     |     |
|                                                                      |                                                                                                             |      |      |     |     |     |     |     |     |     |     |     |
|                                                                      | COs                                                                                                         | PO1  | PO2  | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | P10 | P11 |
|                                                                      | CO1                                                                                                         | 3    | 2    | 2   | 1   |     |     |     |     |     |     |     |
|                                                                      | CO2                                                                                                         | 3    | 2    | 2   | 1   | 1   |     |     |     |     |     |     |
|                                                                      | CO3                                                                                                         | 3    | 2    | 2   | 1   | 1   |     |     |     |     |     |     |
|                                                                      | CO4                                                                                                         | 2    | 1    |     |     | 1   |     |     |     |     |     |     |
|                                                                      | CO5                                                                                                         | 2    | 1    |     |     | 1   |     |     |     |     |     |     |
| COs Vs. PSOs MAPPING                                                 |                                                                                                             |      |      |     |     |     |     |     |     |     |     |     |
|                                                                      | COs                                                                                                         | PSO1 | PSO2 |     |     |     |     |     |     |     |     |     |
|                                                                      | CO1                                                                                                         | 3    |      |     |     |     |     |     |     |     |     |     |
|                                                                      | CO2                                                                                                         | 3    |      |     |     |     |     |     |     |     |     |     |
|                                                                      | CO3                                                                                                         | 3    |      |     |     |     |     |     |     |     |     |     |
|                                                                      | CO4                                                                                                         | 3    |      |     |     |     |     |     |     |     |     |     |
|                                                                      | CO5                                                                                                         | 3    |      |     |     |     |     |     |     |     |     |     |
| COURSE CONTENTS:                                                     |                                                                                                             |      |      |     |     |     |     |     |     |     |     |     |

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| <b>MODULE I</b>                                                                                                                                                                                                                                                                                                                                                                                   | <b>SYNCHRONOUS GENERATOR</b>                                     | <b>12 Hours</b> |
| Constructional details-types of rotors; EMF equation; Armature reaction-phasor diagram; Predetermination of voltage regulation– EMF, MMF, ZPF & ASA methods; Two reaction theory; Slip test; Parallel operation and synchronization to infinite bus bar; Application.                                                                                                                             |                                                                  |                 |
| <b>MODULE II</b>                                                                                                                                                                                                                                                                                                                                                                                  | <b>SYNCHRONOUS MOTOR</b>                                         | <b>12 Hours</b> |
| Principle of operation, torque equation, power input and power developed equations; V and Inverted V curves; Effect of varying load and Excitation; Load test and characteristics; Starting methods; Hunting; Synchronous condenser, Synchronous induction motor, Applications. Simple simulation studies on synchronous motor.                                                                   |                                                                  |                 |
| <b>MODULE III</b>                                                                                                                                                                                                                                                                                                                                                                                 | <b>THREE PHASE INDUCTION MOTOR</b>                               | <b>12 Hours</b> |
| Constructional details- types- principle of operation- rotating magnetic field- slip; Equivalent circuit; Torque equation- Torque-Slip characteristics; No load & Blocked rotor test -Circle diagram, separation of losses; Cogging and crawling; Induction generator; Double cage deep bar induction motor. Introduction to AC drives in industry. Simple simulation studies of induction motor. |                                                                  |                 |
| <b>MODULE IV</b>                                                                                                                                                                                                                                                                                                                                                                                  | <b>STARTING AND SPEED CONTROL OF THREE PHASE INDUCTION MOTOR</b> | <b>12 Hours</b> |
| Need for starter - Types of starters – star delta starter, auto transformer starter, DOL starter- Comparison of starters; Speed control – voltage control, frequency control, pole changing, cascaded connection, v/f control, slip power recovery scheme; Braking methods; Speed control applications in industries; Study on PLC based star delta starter.                                      |                                                                  |                 |
| <b>MODULE V</b>                                                                                                                                                                                                                                                                                                                                                                                   | <b>SINGLE PHASE INDUCTION MOTOR &amp; SPECIAL MOTOR</b>          | <b>12 Hours</b> |
| Principle of single-phase induction motor- Double field revolving theory- Equivalent circuit- Starting methods. Fractional HP motors- Hysteresis motor, Stepper motor; Universal motor; Linear induction motor, D.C. servo motor, PMSM motor, PMDC motor , BLDC motor; Study on PLC based special machines drive.                                                                                 |                                                                  |                 |
| <b>TOTAL: 60 HOURS</b>                                                                                                                                                                                                                                                                                                                                                                            |                                                                  |                 |
| <b>REFERENCES:</b>                                                                                                                                                                                                                                                                                                                                                                                |                                                                  |                 |
| 1. D. P. Kothari and I. J. Nagrath, Electric Machines, Tata McGraw Hill Publishing Company Ltd, 2010.                                                                                                                                                                                                                                                                                             |                                                                  |                 |
| 2. Edward Hughes, Electrical and Electronic Technology, 12 <sup>th</sup> edition, Pearson, 2016.                                                                                                                                                                                                                                                                                                  |                                                                  |                 |
| 3. P. S. Bimbhra, Electrical Machinery, Khanna Publishers, 7th edition, 2011.                                                                                                                                                                                                                                                                                                                     |                                                                  |                 |
| 4. B. L. Theraja and A. K. Theraja, —Text Book of Electrical Technology: AC & DC Machines (Volume- 2),                                                                                                                                                                                                                                                                                            |                                                                  |                 |
| 5. M.N.Bandyopadhyay, Electrical Machines Theory and practice, PHI Learning Pvt. Ltd, New Delhi 2007.                                                                                                                                                                                                                                                                                             |                                                                  |                 |
| 6. Fitzgerald A.E, CharlesKingsley, Stephen. D.Umans, „Electric Machinery“, Tata McGraw-Hill                                                                                                                                                                                                                                                                                                      |                                                                  |                 |
| 7. <a href="http://nptel.ac.in/courses/117106086/">http://nptel.ac.in/courses/117106086/</a>                                                                                                                                                                                                                                                                                                      |                                                                  |                 |



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| <b>MODULE I</b>                                                                                                                                                                                                                                                                                                                      | <b>POWER SEMICONDUCTOR DEVICES</b> | <b>9 Hours</b> |
| Study of switching devices - Power transistors, Power MOSFETs, IGBTs, SCRs, Triacs, GTOs; Static and Dynamic characteristics; Thermal characteristics; losses in the devices.<br><br>SCRs- Two transistor analogy; Snubber circuits; Series and parallel operation of SCRs; Data sheet interpretation of commercially available SCRs |                                    |                |
| <b>MODULE II</b>                                                                                                                                                                                                                                                                                                                     | <b>PHASE CONTROLLED CONVERTERS</b> | <b>9 Hours</b> |
| Principle of phase control; Single phase and three phase half wave and full wave converters with R, RL and RLE loads; Estimation of average and RMS values of load voltage, load current; Performance parameters for converters; Effect of freewheeling diodes; Effect of source inductance; Dual converters; Applications.          |                                    |                |
| <b>MODULE III</b>                                                                                                                                                                                                                                                                                                                    | <b>DC TO DC CONVERTERS</b>         | <b>9 Hours</b> |
| DC choppers: Principle of step-up and step-down choppers, control strategies; Classification of choppers- Single quadrant, two quadrant and four quadrant DC choppers, Buck, Boost and Buck boost converters; SMPS; LUO converter; SEPIC converter; Applications.                                                                    |                                    |                |
| <b>MODULE IV</b>                                                                                                                                                                                                                                                                                                                     | <b>INVERTERS</b>                   | <b>9 Hours</b> |
| Types of inverters; Operation of single phase VSI, three phase VSI (120 and 180-degree modes); Inverter output voltage control; CSIs- Auto sequential CSI; Pulse width modulation techniques- Single, multiple, sinusoidal modulation; Harmonic elimination techniques - SHEPWM; Introduction to MLIs; Applications.                 |                                    |                |
| <b>MODULE V</b>                                                                                                                                                                                                                                                                                                                      | <b>AC TO AC CONVERTERS</b>         | <b>9 Hours</b> |
| AC voltage controllers-Single phase and three phase controllers with R and RL loads; Sequence control of AC regulators – Two stage sequence control - Multistage sequential control.<br><br>Cycloconverters: Step-down and step-up cyclo converters; Introduction to matrix converters; Applications.                                |                                    |                |
| <b>TOTAL: 45 HOURS</b>                                                                                                                                                                                                                                                                                                               |                                    |                |
| <b>REFERENCES:</b>                                                                                                                                                                                                                                                                                                                   |                                    |                |
| 1. Rashid M H, “Power Electronics- Circuits, Devices and Applications”, Prentice Hall of India, New Delhi, 2011.                                                                                                                                                                                                                     |                                    |                |
| 2. P.S.Bimbhra, “Power Electronics”, Khanna Publishers, New Delhi, 2006.                                                                                                                                                                                                                                                             |                                    |                |
| 3. Vedam Subramanyam, “Power Electronics”, New Age International, New Delhi, 1996.                                                                                                                                                                                                                                                   |                                    |                |
| 4. M.D. Singh and K.B. Khanchandani, “Power Electronics”, 2 <sup>nd</sup> Edition, Tata McGraw Hill, New Delhi, 2017.                                                                                                                                                                                                                |                                    |                |
| 5. Ned Mohan, Tore.M.Undeland, William.P.Robbins, “Power Electronics: Converters, Applications and Design”, John                                                                                                                                                                                                                     |                                    |                |
| 6. <a href="https://nptel.ac.in/courses/108/102/108102145/">https://nptel.ac.in/courses/108/102/108102145/</a>                                                                                                                                                                                                                       |                                    |                |

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| 2302EE404                                                            | LINEAR CONTROL SYSTEM                                                                                                |      |     |     | L   | T   | P   | C   |     |     |     |
|                                                                      |                                                                                                                      |      |     |     | 3   | 2   | 0   | 4   |     |     |     |
| PREREQUISITE:                                                        |                                                                                                                      |      |     |     |     |     |     |     |     |     |     |
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|                                                                      | 1. Electric circuit analysis                                                                                         |      |     |     |     |     |     |     |     |     |     |
|                                                                      | 2. Engineering Mathematics                                                                                           |      |     |     |     |     |     |     |     |     |     |
|                                                                      | 3. Engineering Physics                                                                                               |      |     |     |     |     |     |     |     |     |     |
| COURSE OBJECTIVES:                                                   |                                                                                                                      |      |     |     |     |     |     |     |     |     |     |
|                                                                      | 1. To understand the basic components of control systems.                                                            |      |     |     |     |     |     |     |     |     |     |
|                                                                      | 2. To gain the knowledge in time and frequency domain tools for the design and analysis of Feedback control systems. |      |     |     |     |     |     |     |     |     |     |
|                                                                      | 3. To make the students to analyze the stability of linear systems in the frequency domain.                          |      |     |     |     |     |     |     |     |     |     |
|                                                                      | 4. To understand the design of compensator and concepts of state variable analysis.                                  |      |     |     |     |     |     |     |     |     |     |
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| COURSE OUTCOMES:                                                     |                                                                                                                      |      |     |     |     |     |     |     |     |     |     |
| On the successful completion of the course, students will be able to |                                                                                                                      |      |     |     |     |     |     |     |     |     |     |
| CO1                                                                  | Calculate transfer function of various systems using block diagram reduction, signal flow graph technique.           |      |     |     |     |     |     |     |     |     |     |
| CO2                                                                  | Investigate the time response behavior of first and second order system using time domain specification.             |      |     |     |     |     |     |     |     |     |     |
| CO3                                                                  | Analyze the frequency response of open loop transfer function using bode plot and polar plot.                        |      |     |     |     |     |     |     |     |     |     |
| CO4                                                                  | Make use of stability concepts to obtain the desired characteristics                                                 |      |     |     |     |     |     |     |     |     |     |
| CO5                                                                  | Determine the solution for complex control problem.                                                                  |      |     |     |     |     |     |     |     |     |     |
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| COs vs. POs MAPPING                                                  |                                                                                                                      |      |     |     |     |     |     |     |     |     |     |
| COs                                                                  | PO1                                                                                                                  | PO2  | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | P10 | P11 |
| CO1                                                                  | 3                                                                                                                    | 2    | 1   |     | 2   |     |     |     |     |     |     |
| CO2                                                                  | 3                                                                                                                    | 2    | 1   |     | 2   |     |     |     |     |     |     |
| CO3                                                                  | 3                                                                                                                    | 2    | 1   |     | 2   |     |     |     |     |     |     |
| CO4                                                                  | 3                                                                                                                    | 2    | 1   |     | 2   |     |     |     |     |     |     |
| CO5                                                                  | 3                                                                                                                    | 2    | 1   |     | 2   |     |     |     |     |     |     |
| COs vs. PSOs MAPPING                                                 |                                                                                                                      |      |     |     |     |     |     |     |     |     |     |
| COs                                                                  | PSO1                                                                                                                 | PSO2 |     |     |     |     |     |     |     |     |     |
| CO1                                                                  | 3                                                                                                                    |      |     |     |     |     |     |     |     |     |     |
| CO2                                                                  | 3                                                                                                                    |      |     |     |     |     |     |     |     |     |     |
| CO3                                                                  | 3                                                                                                                    |      |     |     |     |     |     |     |     |     |     |
| CO4                                                                  | 3                                                                                                                    |      |     |     |     |     |     |     |     |     |     |
| CO5                                                                  | 3                                                                                                                    |      |     |     |     |     |     |     |     |     |     |
| COURSE CONTENTS:                                                     |                                                                                                                      |      |     |     |     |     |     |     |     |     |     |



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| <b>MODULE I</b>                                                                                                                                                                                                                                                                                               | <b>SYSTEMS AND THEIR REPRESENTATION</b>          | <b>12 Hours</b> |
| Basic elements in control systems; Open and closed loop systems; Transfer function- Electrical and mechanical transfer function models; electrical analogy of mechanical systems, block diagram reduction techniques, signal flow graphs; Simple determination of closed loop transfer function using MATLAB. |                                                  |                 |
| <b>MODULE II</b>                                                                                                                                                                                                                                                                                              | <b>TIME RESPONSE ANALYSIS</b>                    | <b>12 Hours</b> |
| Time response-Time domain specifications, types of test inputs, first and second order system responses; steady state error; type number; Simple time response analysis using MATLAB.                                                                                                                         |                                                  |                 |
| <b>MODULE III</b>                                                                                                                                                                                                                                                                                             | <b>FREQUENCY RESPONSE ANALYSIS</b>               | <b>12Hours</b>  |
| Frequency response –Frequency domain specifications, Bode plot, polar plot, determination of closed loop response from open loop response; Correlation between frequency domain and time domain specifications; Simple frequency response analysis using MATLAB.                                              |                                                  |                 |
| <b>MODULE IV</b>                                                                                                                                                                                                                                                                                              | <b>STABILITY ANALYSIS</b>                        | <b>12 Hours</b> |
| Stability- Concept, relative stability, characteristics equation, Routh Hurwitz criterion, Nyquist stability criterion. Introduction to compensators-Lag, Lead and Lag-lead compensators; simple Nyquist plots using MATLAB.                                                                                  |                                                  |                 |
| <b>MODULE V</b>                                                                                                                                                                                                                                                                                               | <b>DESIGN OF CONTROLLERS AND SYSTEM ANALYSIS</b> | <b>12 Hours</b> |
| Controllers - P, PI and PID controllers, design procedure, tuning of controllers - Root locus concept, Root locus construction; Simple root locus curves in MATLAB.                                                                                                                                           |                                                  |                 |
| <b>TOTAL: 60 HOURS</b>                                                                                                                                                                                                                                                                                        |                                                  |                 |
| <b>REFERENCES:</b>                                                                                                                                                                                                                                                                                            |                                                  |                 |
| 1. Nagarath, I.J. and Gopal, M., “Control Systems Engineering”, New Age International Publishers, 2017.                                                                                                                                                                                                       |                                                  |                 |
| 2. S.K.Bhattacharya, “Control System Engineering”, 3rd Edition, Pearson Publications, New Delhi 2013.                                                                                                                                                                                                         |                                                  |                 |
| 3. K.Ogata, “Modern Control Engineering”, 5th Edition, Pearson Prentice Hall, New Delhi, 2012.                                                                                                                                                                                                                |                                                  |                 |
| 4. Richard C.Dorf and Robert H. Bishop, “Modern Control Systems”, 12th Edition, Pearson Prentice Hall, 2012. Benjamin C. Kuo, “Automatic Control Systems”, Wiley, 2014.                                                                                                                                       |                                                  |                 |
| 5. Benjamin C. Kuo, “Automatic Control Systems”, Wiley, 2014.                                                                                                                                                                                                                                                 |                                                  |                 |
| 6. <a href="https://nptel.ac.in/courses/107106081">https://nptel.ac.in/courses/107106081</a>                                                                                                                                                                                                                  |                                                  |                 |
| 7. <a href="https://cosmolearning.org/courses/control-engineering/">https://cosmolearning.org/courses/control-engineering/</a>                                                                                                                                                                                |                                                  |                 |
| 8. <a href="https://library.villanova.edu/Find/Record/1437935/TOC">https://library.villanova.edu/Find/Record/1437935/TOC</a>                                                                                                                                                                                  |                                                  |                 |

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| 2302EE405                                                            | LINEAR INTEGRATED CIRCUITS                                                                                                       |     |     |     | L   | T   | P   | C   |     |     |     |
|                                                                      |                                                                                                                                  |     |     |     | 3   | 0   | 2   | 4   |     |     |     |
| PREREQUISITE:                                                        |                                                                                                                                  |     |     |     |     |     |     |     |     |     |     |
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|                                                                      | 1. Electric circuit Analysis                                                                                                     |     |     |     |     |     |     |     |     |     |     |
|                                                                      | 2. Analog Electronics                                                                                                            |     |     |     |     |     |     |     |     |     |     |
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| COURSE OBJECTIVES:                                                   |                                                                                                                                  |     |     |     |     |     |     |     |     |     |     |
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| 1                                                                    | To understand the fundamentals and fabrication of ICs.                                                                           |     |     |     |     |     |     |     |     |     |     |
| 2                                                                    | To explain the functions, characteristics and applications of op. amp.                                                           |     |     |     |     |     |     |     |     |     |     |
| 3                                                                    | To describe operation of signal converters, special function ICs and voltage regulators                                          |     |     |     |     |     |     |     |     |     |     |
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| COURSE OUTCOMES:                                                     |                                                                                                                                  |     |     |     |     |     |     |     |     |     |     |
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| On the successful completion of the course, students will be able to |                                                                                                                                  |     |     |     |     |     |     |     |     |     |     |
| CO1                                                                  | Explain the fundamentals of IC technology and fabrication procedure for diode, capacitance, resistance, FET and typical circuits |     |     |     |     |     |     |     |     |     |     |
| CO2                                                                  | Describe the functional block diagram, performance parameters and frequency compensation techniques of operational amplifier     |     |     |     |     |     |     |     |     |     |     |
| CO3                                                                  | Construct analog circuits using operational amplifier for linear and non-linear applications.                                    |     |     |     |     |     |     |     |     |     |     |
| CO4                                                                  | Build signal conversion circuits and filters using operational amplifier.                                                        |     |     |     |     |     |     |     |     |     |     |
| CO5                                                                  | Design simple analog circuits for the given application using timer, VCO, PLL and voltage regulator ICs                          |     |     |     |     |     |     |     |     |     |     |
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| COs vs. POs MAPPING                                                  |                                                                                                                                  |     |     |     |     |     |     |     |     |     |     |
| COs                                                                  | PO1                                                                                                                              | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | P10 | P11 |
| CO1                                                                  | 2                                                                                                                                | 1   |     |     |     |     |     |     |     |     |     |
| CO2                                                                  | 2                                                                                                                                | 2   | 1   |     |     |     |     |     |     |     |     |
| CO3                                                                  | 3                                                                                                                                | 2   | 1   |     |     |     |     |     |     |     |     |
| CO4                                                                  | 3                                                                                                                                | 2   | 1   |     |     |     |     |     |     |     |     |
| CO5                                                                  | 3                                                                                                                                | 2   | 1   |     |     |     |     |     |     |     |     |
| COs vs. PSOs MAPPING                                                 |                                                                                                                                  |     |     |     |     |     |     |     |     |     |     |
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| <b>MODULE I</b>                                                                                                                                                                                                                                                                                                                                                              | <b>FABRICATION OF INTEGRATED CIRCUITS</b>     | <b>9 Hours</b> |
| IC classification; Fundamental of monolithic IC technology; Epitaxial growth- masking and etching- diffusion of impurities; realization of monolithic ICs and packaging; Fabrication of diodes, capacitance, resistance and FET.                                                                                                                                             |                                               |                |
| <b>MODULE II</b>                                                                                                                                                                                                                                                                                                                                                             | <b>OPERATIONAL AMPLIFIER</b>                  | <b>9 Hours</b> |
| Op-Amp- functional block diagram, ideal and practical op-amp; IC741- Pin diagram- Features- Interpretation of IC 741 datasheet; Characteristics- CMRR, open loop gain, slew rate; transfer characteristics, input bias and output offset voltage, offset compensation techniques; frequency response characterization, frequency compensation.                               |                                               |                |
| <b>MODULE III</b>                                                                                                                                                                                                                                                                                                                                                            | <b>APPLICATIONS OF OPERATIONAL AMPLIFIERS</b> | <b>9 Hours</b> |
| Inverting and non-inverting amplifiers- voltage follower- summing amplifier- differential amplifier- instrumentation amplifier; Comparators; Integrator and differentiator; Precision rectifier; Logarithmic and anti logarithmic amplifiers. Sinusoidal oscillators - phase shift, Wein bridge & Hartley; sample and hold circuit; clipper and clamper; Schmitt trigger.    |                                               |                |
| <b>MODULE IV</b>                                                                                                                                                                                                                                                                                                                                                             | <b>SIGNAL CONVERSION CIRCUITS</b>             | <b>9 Hours</b> |
| V/F and F/V converters; V/I and I/V converter; D/A converter - weighted resistor type, R-2R ladder type, inverted R-2R, comparison; A/D converters- flash type, successive approximation type, single slope type, dual slope type, A/D converter using voltage-to-time conversion, comparison.                                                                               |                                               |                |
| <b>MODULE V</b>                                                                                                                                                                                                                                                                                                                                                              | <b>SPECIAL FUNCTION INTEGRATED CIRCUITS</b>   | <b>9 Hours</b> |
| 555 Timer - functional block diagram and description- astable, monostable and bistable operations; 566 voltage controlled oscillator; 565 PLL - functional block diagram, principle of operation, characteristics; IC voltage regulators – regulation, need for voltage regulation; LM78XX, 79XX fixed voltage regulators; LM 317 & LM723; Interpretation of IC data sheets. |                                               |                |
| <b>TOTAL: 45 HOURS</b>                                                                                                                                                                                                                                                                                                                                                       |                                               |                |
| <b>LIST OF EXPERIMENTS:</b>                                                                                                                                                                                                                                                                                                                                                  |                                               |                |
| 1. Inverting and non-inverting amplifiers using op-amp.                                                                                                                                                                                                                                                                                                                      |                                               |                |
| 2. Adder, subtractor and comparators using op-amp.                                                                                                                                                                                                                                                                                                                           |                                               |                |
| 3. Weighted resistor type DAC and R-2R ladder type DAC                                                                                                                                                                                                                                                                                                                       |                                               |                |
| 4. Sinusoidal oscillators using op-amps.                                                                                                                                                                                                                                                                                                                                     |                                               |                |
| 5. Astable and monostable multivibrators using NE555 timer.                                                                                                                                                                                                                                                                                                                  |                                               |                |
| <b>TOTAL: 15 HOURS</b>                                                                                                                                                                                                                                                                                                                                                       |                                               |                |
| <b>TOTAL:(45+15) 60 HOURS</b>                                                                                                                                                                                                                                                                                                                                                |                                               |                |
| <b>REFERENCES:</b>                                                                                                                                                                                                                                                                                                                                                           |                                               |                |
| 1. D Roy Choudhury and SheilB.Jani, “Linear Integrated Circuits” 4th Edition, New Age International, New Delhi, 2014.                                                                                                                                                                                                                                                        |                                               |                |
| 2. S Salivahanan and V S KanchanaBhaaskaran, “Linear Integrated Circuits”, 2nd Edition, McGraw-Hill Education, 2014.                                                                                                                                                                                                                                                         |                                               |                |
| 3. RamakantA.Gayakward, “Op-amps and Linear Integrated Circuits”, 4thEdition, PHI Learnings, 2003.                                                                                                                                                                                                                                                                           |                                               |                |
| 4. B Somanathan Nair, “Linear Integrated Circuits: Analysis, Design and Applications”, Wiley, 2009.                                                                                                                                                                                                                                                                          |                                               |                |
| 5. Floyd and Buchla, “Fundamentals of Analog Circuits”, Pearson, 2013.                                                                                                                                                                                                                                                                                                       |                                               |                |
| 6. James M. Fiore, “Operational Amplifiers & Linear Integrated Circuits: Theory and Application / 3E”, 2018.                                                                                                                                                                                                                                                                 |                                               |                |
| 7. Microelectronic circuits-by A.S.Sedra and K.C.Smith                                                                                                                                                                                                                                                                                                                       |                                               |                |
| 8. <a href="http://nptel.ac.in/courses/117107094/">http://nptel.ac.in/courses/117107094/</a>                                                                                                                                                                                                                                                                                 |                                               |                |

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| <b>2301HSX01</b> | <b>UNIVERSAL HUMAN VALUES AND ETHICS</b> | <b>L</b> | <b>T</b> | <b>P</b> | <b>C</b> |
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**PREREQUISITE:**

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|  | 1. Professional Ethics |
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**COURSE OBJECTIVES:**

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| 1 | Reinstate India's rich cultural legacy and human values of which we are the custodians.                     |
| 2 | Focus on professional ethics, which help citizens to discern desirable and undesirable actions.             |
| 3 | Re-emphasize constitutional values, universal values, and holistic education to create integrated citizens. |
| 4 | Lay down broader guidelines of human values and ethics for internal and external stakeholders.              |

**COURSE OUTCOMES:**

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| On the successful completion of the course, students will be able to |                                                                                                                                                                                                    |
| <b>CO1</b>                                                           | Apply critical thinking skills to solve problems and make informed decisions in various contexts.                                                                                                  |
| <b>CO2</b>                                                           | Analyze the principles of effective self - governance and evaluate their implementation in different scenarios.                                                                                    |
| <b>CO3</b>                                                           | Understand the importance of a fair and transparent system of rewards and reprimand in constitution of India and apply these principles in real-world situations.                                  |
| <b>CO4</b>                                                           | Analyze the role of an individual to develop social reliability and create awareness about their assertiveness and self - confidence.                                                              |
| <b>CO5</b>                                                           | Understand the knowledge of inner qualities and instruments of one's self in stress management and analyze the effects of meditation in one's physical, mental, social and spiritual well - being. |

**COs Vs POs MAPPING:**

| COs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | 2   | 1   | -   | -   | 1   | 3   | 1   | 2   | -    | 2    |
| CO2 | 3   | 3   | 2   | -   | -   | 1   | 2   | 1   | 2   | -    | 2    |
| CO3 | 2   | 1   | -   | -   | -   | 2   | 2   | 1   | 2   | -    | 1    |
| CO4 | 3   | 3   | 2   | -   | -   | 1   | 2   | 2   | 2   | -    | 1    |
| CO5 | 2   | 1   | -   | -   | -   | 2   | 3   | 1   | 2   | -    | 1    |

**COs Vs. PSOs MAPPING**

| COs | PSO1 | PSO2 |
|-----|------|------|
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| CO2 | -    | -    |
| CO3 | -    | -    |
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| <b>COURSE CONTENTS:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                     |                 |
| <b>MODULE I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Introduction to Indian Ethos</b>                 | <b>08 Hours</b> |
| Meaning of ethos and cultural essence of India – Scriptures as the base of the Indian Knowledge System (IKS) – Integrating the two methodologies: interiorization process for self-exploration, and exterior scientific pursuit for the prosperity of world – The Law of Karma and Nishkama Karma (The Law of action and selfless action).<br><b>Practical:</b> Five hours of Yoga practice per week, Ethics through Music and Indian Poetry, Community Engagement.                                                                                                      |                                                     |                 |
| <b>MODULE II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Human Values and Ethics</b>                      | <b>09 Hours</b> |
| Knowing the Self and the universal values that we stand for - This is self-enquiry & self-discovery – Background conversations and deep listening - recognizing the assumptions that we make - the biases we have - and the implications for ethical action – Self-identity: distinguishing and embracing oneself (and others) four profiles (inner-potential, social, professional, personality) – Distinguish ideology, perspectives beliefs from embodying values.<br><b>Practical:</b> Self-discovery, self-enquiry and Mindfulness, Yama & Niyama of Ashthang Yoga. |                                                     |                 |
| <b>MODULE III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Constitutional Values and Global Citizenship</b> | <b>09 Hours</b> |
| Values embedded in the Preamble of the Indian Constitution Integration of Human Rights and duties – Directive principles and responsibilities as citizens of India – Sensibility and responsibilities towards global environment, Loksangraha and Vasudhaiva Kutumbakam.<br><b>Practical:</b> Debates and Theatre on diversity and plurality, research on similarities and differences in the ethos of different countries.                                                                                                                                              |                                                     |                 |
| <b>MODULE IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Values and Skills for Youth</b>                  | <b>09 Hours</b> |
| Designing to make a difference through strategies using the Conscious Full Spectrum Response model – Listening for commitment behind complaints to transform contentious arguments and create a space for listening and change – Distinguishing judgement from discernment – Being assertive and confident (assertiveness incorporates self-confidence).<br><b>Practical :</b> Development of concentration among students through music, fine arts, mathematics, sports, yoga and mindfulness                                                                           |                                                     |                 |
| <b>MODULE V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Integrated Personality and Well-being</b>        | <b>10 Hours</b> |
| The three gunas (qualities of sattva—purity and harmony, rajas —activity and passion, tamas — darkness and chaos), the four antah-karanas (inner instruments), and panchkosha (five sheaths) – Stress management: meditated personality and agitated personality – Oneness, non-duality, and equanimity – Physical, mental, social, and spiritual well-being.<br><b>Practical :</b> Talks on importance of the Ayurvedic concept of well-being and nutrition, sports activities                                                                                          |                                                     |                 |
| <b>TOTAL: 45 HOURS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                     |                 |
| <b>TEXT BOOKS:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                     |                 |
| 1. R R Gaur, R Sangal, G P Bagaria, 2009, A Foundation Course in Human Values and Professional Ethics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                     |                 |
| <b>REFERENCES:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                     |                 |
| 1. Blanchard, Kenneth and Peale, Norman Vincent. 1988. <i>The Power of Ethical Management</i> . New York: William Morrow and Company, Inc.                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                     |                 |
| 2. Gandhi, Mohandas Karamchand. 1971. <i>Pathway to God</i> compiled by MS Deshpande. Ahmedabad: Navajivan Mudranalaya, Navjivan Trust.                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                     |                 |
| 3. <a href="https://fdp-si.aicte-india.org/UHV-II%20Class%20Note.php">https://fdp-si.aicte-india.org/UHV-II%20Class%20Note.php</a>                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                     |                 |

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| 2301MC401                                                                                                                                                                                                                                                                                 | DESIGN THINKING FOR INNOVATION                                                                                                                                                                                                                                                                                     |  | L | T | P        | C |
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| <b>PREREQUISITE:</b>                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
|                                                                                                                                                                                                                                                                                           | The course assumes no prior skill or background in design, art, engineering, or prototyping. It is open to all undergraduates and graduate students with an interest in learning design thinking, and is especially recommended for those students planning social-venture and other kinds of design interventions |  |   |   |          |   |
| <b>COURSE OBJECTIVES:</b>                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| 1.                                                                                                                                                                                                                                                                                        | Understand the terminology and conceptual models used in design disciplines                                                                                                                                                                                                                                        |  |   |   |          |   |
| 2.                                                                                                                                                                                                                                                                                        | Understand how teaching and learning occurs in the design process                                                                                                                                                                                                                                                  |  |   |   |          |   |
| 3.                                                                                                                                                                                                                                                                                        | Recognize the ethical and social dilemmas and obligations of the practice of design                                                                                                                                                                                                                                |  |   |   |          |   |
| 4.                                                                                                                                                                                                                                                                                        | Diagnose common adoption barriers in individuals, groups and organizations.                                                                                                                                                                                                                                        |  |   |   |          |   |
| 5.                                                                                                                                                                                                                                                                                        | Develop a design theory from independent and qualitative research and observations                                                                                                                                                                                                                                 |  |   |   |          |   |
| 6.                                                                                                                                                                                                                                                                                        | Participate in and lead innovation in creative and collaborative settings                                                                                                                                                                                                                                          |  |   |   |          |   |
| 7.                                                                                                                                                                                                                                                                                        | Undertake complex and unstructured problem-solving challenges in unfamiliar domains                                                                                                                                                                                                                                |  |   |   |          |   |
| <b>COURSE OUTCOMES:</b>                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| Upon successful completion of the course, students will be able to                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| CO1                                                                                                                                                                                                                                                                                       | Describe Key Concepts and basics of Design Thinking Principles                                                                                                                                                                                                                                                     |  |   |   |          |   |
| CO2                                                                                                                                                                                                                                                                                       | Elaborate the Design Thinking Approach through IDEO's method & Customer Journey Maps                                                                                                                                                                                                                               |  |   |   |          |   |
| CO3                                                                                                                                                                                                                                                                                       | Conduct user interviews and synthesize learning's to uncover insights and identify opportunities for innovation                                                                                                                                                                                                    |  |   |   |          |   |
| CO4                                                                                                                                                                                                                                                                                       | Develop Design Driven Innovative Solutions to Real-world Problems                                                                                                                                                                                                                                                  |  |   |   |          |   |
| <b>COURSE CONTENTS:</b>                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| Module-I                                                                                                                                                                                                                                                                                  | Introduction to Design Thinking                                                                                                                                                                                                                                                                                    |  |   |   | 8 Hours  |   |
| Human Centered Design, Why Design Thinking, 5-Step Design Thinking Process, Applications, Creative Confidence, The culture of Innovation                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| Module-II                                                                                                                                                                                                                                                                                 | Design Thinking Approach                                                                                                                                                                                                                                                                                           |  |   |   | 12 Hours |   |
| IDEO's method of Design Thinking, Divergent Thinking & Innovation Funnel, Customer Journey Maps to uncover Innovation Opportunities, Case Study : Turing Creative Ideas into Viable Companies                                                                                             |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| Module-III                                                                                                                                                                                                                                                                                | Exploring Design Thinking Tool Kit                                                                                                                                                                                                                                                                                 |  |   |   | 5 Hours  |   |
| Discovery, Interpretation, Ideation, Experimentation, Evolution                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| Module-IV                                                                                                                                                                                                                                                                                 | Design Challenge Project : Phase-1                                                                                                                                                                                                                                                                                 |  |   |   | 5 Hours  |   |
| Define a Challenge, Project Plan, How Might We statements, Project Timeline, Project Checklist                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| Module-V                                                                                                                                                                                                                                                                                  | Design Challenge Project : Phase-2                                                                                                                                                                                                                                                                                 |  |   |   | 15 Hours |   |
| Discovery – Understand the Challenge, Prepare Research, Gather Inspiration, Interpretation – Tell Stories, Search for meaning, Frame Opportunities, Ideation – Generate Ideas, Refine Ideas, Experimentation – Make Prototypes, Get Feedback, Evolution – Track Learning's, Engage Others |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| <b>TOTAL: 45 HOURS</b>                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| <b>FURTHER READING:</b>                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| 1. Design for Social Impact : How to by IDEO.org                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| 2. Design Thinking Tool Kit by IDEO.org                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| 3. The Field guide to Human Centered Design by IDEO.org                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| <b>REFERENCES:</b>                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| 1. Creative Confidence: Unleashing the Creative Potential Within Us All Book by David M. Kelley and Tom Kelley, 2013                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |
| 2. Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                    |  |   |   |          |   |

|           |                                     |   |   |   |   |
|-----------|-------------------------------------|---|---|---|---|
| 2302EE451 | ELECTRICAL MACHINES – II LABORATORY | L | T | P | C |
|           |                                     | 0 | 0 | 2 | 1 |

**PREREQUISITE:**

1. Electrical Machines –I Laboratory

**COURSE OBJECTIVES:**

1. To know the performance characteristics of induction motors.
2. To compare various regulation methods of Synchronous machines.
3. To study the characteristics of brushless DC motor.

**COURSE OUTCOMES:**

On the successful completion of the course, students will be able to

- |            |                                                                                                                                          |
|------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CO1</b> | Investigate various regulation methods of synchronous machines by conducting OC and SC test                                              |
| <b>CO2</b> | Compute the performance characteristics of single phase / three-phase induction motor by conducting Load, no load and blocked rotor test |
| <b>CO3</b> | Identify the performance characteristics of three-phase/single -phase induction motor by conducting OC and SC test                       |
| <b>CO4</b> | Compute the load sharing of synchronous machines with bus bar                                                                            |
| <b>CO5</b> | Analyze the behavior of special machine Drives using PLC                                                                                 |

**COs vs. POs MAPPING**

| COs        | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | P10 | P11 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| <b>CO1</b> | 3   | 3   | 3   | 3   |     |     |     |     | 2   | 2   | 2   |
| <b>CO2</b> | 3   | 3   | 3   | 3   |     |     |     |     | 2   | 2   | 2   |
| <b>CO3</b> | 3   | 3   | 3   | 3   |     |     |     |     | 2   | 2   | 2   |
| <b>CO4</b> | 3   | 3   | 3   | 3   |     |     |     |     | 2   | 2   | 2   |
| <b>CO5</b> | 3   | 3   | 3   | 3   | 2   |     |     |     | 2   | 2   | 2   |

**COs vs. PSOs MAPPING**

| COs        | PSO1 | PSO2 |
|------------|------|------|
| <b>CO1</b> | 3    |      |
| <b>CO2</b> | 3    |      |
| <b>CO3</b> | 3    |      |
| <b>CO4</b> | 3    |      |
| <b>CO5</b> | 3    |      |

**COURSE CONTENTS:**

1. Voltage regulation of an alternator by EMF and MMF method.

|                                                                                                                                               |                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 2.                                                                                                                                            | Voltage regulation of an alternator by ZPF and ASA method.                                                 |
| 3.                                                                                                                                            | Determination of $X_d$ , $X_q$ and regulation of a salient pole alternator.                                |
| 4.                                                                                                                                            | Parallel operation of alternators/ synchronization of alternator with infinite bus bar.                    |
| 5.                                                                                                                                            | No load, blocked rotor test on three phase induction motor<br>a) Circle Diagram.<br>b) Equivalent Circuit. |
| 6.                                                                                                                                            | Load Test on three-phase induction motor.                                                                  |
| 7.                                                                                                                                            | Separation of no load losses of three-phase induction motor.                                               |
| 8.                                                                                                                                            | No load, blocked rotor test on single phase induction motor – Equivalent Circuit.                          |
| 9.                                                                                                                                            | Load Test on single-phase induction motor.                                                                 |
| 10.                                                                                                                                           | PLC based Special Machine Drive (BLDC).                                                                    |
| 11.                                                                                                                                           | Demonstration on PLC based Star – Delta starter.                                                           |
| <b>TOTAL: 30 HOURS</b>                                                                                                                        |                                                                                                            |
| <b>REFERENCES:</b>                                                                                                                            |                                                                                                            |
| 1. Kothari.D.P&Umre.B.S “Laboratory manual for electrical machines”, I.K international Publishing House (P)Ltd. 2 <sup>nd</sup> Edition, 2017 |                                                                                                            |



|                                                                                                                   |                                                                                         |      |     |     |     |     |     |     |     |     |     |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 2302EE452                                                                                                         | POWER ELECTRONICS AND DRIVES LABORATORY                                                 |      |     |     | L   | T   | P   | C   |     |     |     |
|                                                                                                                   |                                                                                         |      |     |     | 0   | 0   | 2   | 1   |     |     |     |
| PREREQUISITE:                                                                                                     |                                                                                         |      |     |     |     |     |     |     |     |     |     |
|                                                                                                                   | 1. Power Electronics                                                                    |      |     |     |     |     |     |     |     |     |     |
|                                                                                                                   | 2. Electrical Machinery – I &II                                                         |      |     |     |     |     |     |     |     |     |     |
| COURSE OBJECTIVES:                                                                                                |                                                                                         |      |     |     |     |     |     |     |     |     |     |
| 1                                                                                                                 | To determine the characteristics of power electronic devices.                           |      |     |     |     |     |     |     |     |     |     |
| 2                                                                                                                 | To design a power converter for electrical drives.                                      |      |     |     |     |     |     |     |     |     |     |
| 3                                                                                                                 | To analyze the performance of power converter fed drives.                               |      |     |     |     |     |     |     |     |     |     |
| COURSE OUTCOMES:                                                                                                  |                                                                                         |      |     |     |     |     |     |     |     |     |     |
|                                                                                                                   | After completion of the course, Student will be able to                                 |      |     |     |     |     |     |     |     |     |     |
| CO1                                                                                                               | Construct experiments on power electronic component for obtaining characteristics curve |      |     |     |     |     |     |     |     |     |     |
| CO2                                                                                                               | Design and test AC-DC and AC-AC converters                                              |      |     |     |     |     |     |     |     |     |     |
| CO3                                                                                                               | Design and test chopper and inverter circuits                                           |      |     |     |     |     |     |     |     |     |     |
| CO4                                                                                                               | Design pulse generation circuits for power electronic converters                        |      |     |     |     |     |     |     |     |     |     |
| CO5                                                                                                               | Evaluate the performance of power converters through simulation                         |      |     |     |     |     |     |     |     |     |     |
| COs vs. POs MAPPING                                                                                               |                                                                                         |      |     |     |     |     |     |     |     |     |     |
| COs                                                                                                               | PO1                                                                                     | PO2  | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | P10 | P11 |
| CO1                                                                                                               | 3                                                                                       | 3    | 3   | 3   |     |     |     |     | 2   | 2   | 2   |
| CO2                                                                                                               | 3                                                                                       | 3    | 3   | 3   |     |     |     |     | 2   | 2   | 2   |
| CO3                                                                                                               | 3                                                                                       | 3    | 3   | 3   |     |     |     |     | 2   | 2   | 2   |
| CO4                                                                                                               | 3                                                                                       | 3    | 3   | 3   |     |     |     |     | 2   | 2   | 2   |
| CO5                                                                                                               | 3                                                                                       | 3    | 3   | 3   | 3   |     |     |     | 2   | 2   | 2   |
| COs vs. PSOs MAPPING                                                                                              |                                                                                         |      |     |     |     |     |     |     |     |     |     |
| COs                                                                                                               | PSO1                                                                                    | PSO2 |     |     |     |     |     |     |     |     |     |
| CO1                                                                                                               |                                                                                         | 3    |     |     |     |     |     |     |     |     |     |
| CO2                                                                                                               |                                                                                         | 3    |     |     |     |     |     |     |     |     |     |
| CO3                                                                                                               |                                                                                         | 3    |     |     |     |     |     |     |     |     |     |
| CO4                                                                                                               |                                                                                         | 3    |     |     |     |     |     |     |     |     |     |
| CO5                                                                                                               |                                                                                         | 3    |     |     |     |     |     |     |     |     |     |
| LIST OF EXPERIMENTS                                                                                               |                                                                                         |      |     |     |     |     |     |     |     |     |     |
| 1. Switching characteristics of SCR, TRAIC and IGBT.                                                              |                                                                                         |      |     |     |     |     |     |     |     |     |     |
| 2. Performance evaluation of single phase half wave and full wave AC – DC controlled converter drive application. |                                                                                         |      |     |     |     |     |     |     |     |     |     |
| 3. Performance evaluation of three phase half wave and full wave AC – DC controlled converter drive application.  |                                                                                         |      |     |     |     |     |     |     |     |     |     |
| 4. Step down, step up and Multi-quadrant MOSFET-based choppers.                                                   |                                                                                         |      |     |     |     |     |     |     |     |     |     |
| 5. IGBT-based single phase PWM inverter.                                                                          |                                                                                         |      |     |     |     |     |     |     |     |     |     |
| 6. IGBT-based three-phase PWM inverter fed three phase AC Motor.                                                  |                                                                                         |      |     |     |     |     |     |     |     |     |     |
| 7. Gate Pulse Generation using digital programming circuits for DC – DC converter & single-phase inverter.        |                                                                                         |      |     |     |     |     |     |     |     |     |     |

|                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------|
| 8. Performance evaluation of single phase AC voltage controller.                                                    |
| 9. V/F control of induction motor drive.                                                                            |
| 10. Simulation of PE circuits (1 $\Phi$ & 3 $\Phi$ semi converter, full converters, DC-DC converters and inverter). |
| <b>TOTAL:</b> <b>30 Hours</b>                                                                                       |
| <b>REFERENCES:</b>                                                                                                  |
| 1. K. Nandakumar, R. Anandaraj, “Power Electronics and Drives Laboratory Manual”, 2018                              |
| 2. Krishnan. R, “Electric Motor and Drives Modeling, Analysis and Control”, Prentice Hall of India, 2001.           |
| 3. Pillai, S.K., “A First Course on Electrical Drives”, Wiley Eastern Limited, 1993.                                |